New CP Backend API

OpenAPI 3.0.3 | Version 1.0.0

New CP backend API contract for authentication, tenant-scoped IAM, and approved service-instance workflows.

Base Server URL	http://localhost:3001 (Local New CP backend)
Authentication	HTTP Bearer Token (BearerAuth — opaque session token or Google JWT)
Contract Status	APPROVED_QB8_QB15_IMPLEMENTATION_TARGET; F07_AND_LIVE_API_UNVERIFIED; NOT_FINAL
F07 Target Mapping	Column: orders.order_items.billed_to_contact_id | Origin: orders.order_items.billed_party_origin
Independent Billability	Column: orders.order_items.is_billable (V011 existing NOT NULL DEFAULT true; independent of F07 V204)
Concurrency & Locks	Aggregate ETag + required If-Match (412 stale / 428 missing); financial lock on charged/invoiced lines

Endpoints Overview

Method	Endpoint Path	Tag	Operation ID & Summary
POST	`/api/v1/auth/login`	Auth	**loginWithGoogle**: Authenticate with a Google ID token
GET	`/health`	System	**getHealth**: Liveness/readiness probe
GET	`/api/v1/catalog/services`	Catalog	**listDemoCatalogServices**: List authorized sellable service options (tenant scoped)
GET	`/api/v1/catalog/services/{serviceUuid}/form-schema`	Catalog	**getServiceFormSchema**: Get actual tenant catalog field and condition metadata
GET	`/api/v1/orders/{orderUuid}/services`	Service Instances	**listOrderServiceItems**: List independent service instances on an authorized order
POST	`/api/v1/orders/{orderUuid}/services`	Service Instances	**createOrderServiceItem**: Create a new independent service instance on an order
GET	`/api/v1/services/{serviceItemUuid}`	Service Instances	**getServiceItem**: Get a single service instance with ETag aggregate revision
PATCH	`/api/v1/services/{serviceItemUuid}`	Service Instances	**patchServiceItem**: Update a service instance with If-Match concurrency guard
GET	`/api/v1/services/{serviceItemUuid}/available-transitions`	Workflow	**listServiceTransitions**: List currently authorized manual workflow transitions
POST	`/api/v1/services/{serviceItemUuid}/transitions`	Workflow	**advanceServiceWorkflow**: Execute an atomic workflow state transition with If-Match guard
GET	`/api/v1/services/{serviceItemUuid}/history`	Workflow	**getServiceHistory**: Get status and workflow transition history for a service instance
GET	`/api/v1/orders/{orderUuid}/eligible-billed-parties`	Service Instances	**listEligibleOrderItemBilledParties**: List eligible service-level billed parties for one authorized order
GET	`/api/v1/catalog/hold-reasons`	Catalog, Lookups	**listEligibleHoldReasons**: Return configured hold reasons; temporary synthetic choices are nonproduction only
GET	`/api/v1/lookups/zero-charge-reasons`	Lookups	**listZeroChargeReasons**: List active zero charge reasons
GET	`/api/v1/lookups/hold-reasons`	Lookups	**listHoldReasons**: List active hold reasons (lookup alias for /api/v1/catalog/hold-reasons)
GET	`/api/v1/auth/me`	Auth	**getCurrentAuthContext**: Get the current authenticated user and tenant authorization context

Method	Endpoint Path	Tag	Operation ID & Summary
POST	`/api/v1/auth/logout`	Auth	**logoutApplicationSession**: Revoke the current application session
GET	`/api/v1/iam/users`	IAM/Admin	**listIamUsers**: List tenant-scoped employee identities for IAM administration
GET	`/api/v1/iam/roles`	IAM/Admin	**listIamRoles**: List tenant-scoped IAM roles and capabilities
POST	`/api/v1/iam/users/{userId}/roles`	IAM/Admin	**assignIamUserRole**: Assign a tenant role to an employee

Detailed Endpoint Specifications

[POST] /api/v1/auth/login

Operation ID: loginWithGoogle | **Tag:** Auth | **Summary:** Authenticate with a Google ID token

Google verifies identity only. Tenant and role authorization is resolved by the backend IAM store; Google Groups are not used for authorization.

Request Body: (Required: Yes) Schema: Ref: AuthLoginRequest

Responses:

Status	Description & Headers	Response Schema
200	Authenticated identity with either an authorized application session or Pending Access	oneOf: Ref: AuthorizedLoginResponse | Ref: PendingAccessLoginResponse
400	Missing or invalid request body	Ref: AuthErrorResponse
401	Google authentication failed or provisioning eligibility failed	Ref: AuthErrorResponse
403	Configured tenant is unavailable or not authorized	Ref: AuthErrorResponse

[GET] /health

Operation ID: getHealth | **Tag:** System | **Summary:** Liveness/readiness probe

Responses:

Status	Description & Headers	Response Schema
200	Service is healthy	object

[GET] /api/v1/catalog/services

Operation ID: listDemoCatalogServices | **Tag:** Catalog | **Summary:** List authorized sellable service options (tenant scoped)

Parameters:

Name	In	Required	Type	Description
`categoryCode`	query	No	string	-

Responses:

Status	Description & Headers	Response Schema
200	Success	array
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	Ref: Problem

[GET] /api/v1/catalog/services/{serviceUuid}/form-schema

Operation ID: getServiceFormSchema | **Tag:** Catalog | **Summary:** Get actual tenant catalog field and condition metadata

Parameters:

Name	In	Required	Type	Description
`serviceUuid`	path	**Yes**	string (uuid)	-

Responses:

Status	Description & Headers	Response Schema
200	Success	Ref: FormDefinition
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	Ref: Problem

[GET] /api/v1/orders/{orderUuid}/services

Operation ID: listOrderServiceItems | **Tag:** Service Instances | **Summary:** List independent service instances on an authorized order

Parameters:

Name	In	Required	Type	Description
`orderUuid`	path	**Yes**	string (uuid)	-

Responses:

Status	Description & Headers	Response Schema
200	Success	array
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	Ref: Problem

[POST] /api/v1/orders/{orderUuid}/services

Operation ID: createOrderServiceItem | **Tag:** Service Instances | **Summary:** Create a new independent service instance on an order

Create a new independent service-instance UUID even when same sellable item already exists. Server enforces field rules, authorization and tenant-scoped transaction. Idempotency-Key proposal requires explicit policy confirmation; intentional second submissions use distinct keys.

Parameters:

Name	In	Required	Type	Description
`orderUuid`	path	**Yes**	string (uuid)	-
`Idempotency-Key`	header	No	string	Proposed retry protection; exact retention/replay policy is a technical decision.

Request Body: (Required: Yes) Schema: Ref: CreateServiceItem

Responses:

Status	Description & Headers	Response Schema
201	Created service item Header **ETag** (Initial opaque revision of the created service aggregate.)	Ref: ServiceItem
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	Ref: Problem

[GET] /api/v1/services/{serviceItemUuid}

Operation ID: getServiceItem | **Tag:** Service Instances | **Summary:** Get a single service instance with ETag aggregate revision

Return ETag covering the entire editable service state. Implementation must update revision on child field/workflow writes, not only order_items.updated_at.

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	string (uuid)	-

Responses:

Status	Description & Headers	Response Schema
200	Success Header **ETag** (Opaque revision of the entire editable service aggregate (including fields/workflow). Store and send on next write.)	Ref: ServiceItem
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	Ref: Problem

[PATCH] /api/v1/services/{serviceItemUuid}

Operation ID: patchServiceItem | **Tag:** Service Instances | **Summary:** Update a service instance with If-Match concurrency guard

Requires If-Match; approved CP authorized unitPrice overrides must not be silently replaced with contract rate. Preserve zero-charge reason required at final line_total zero and financial locks.

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	string (uuid)	-
`If-Match`	header	**Yes**	string	ETag of loaded service aggregate. Required on mutations. 412 if stale, 428 if absent.

Request Body: (Required: Yes) Schema: Ref: PatchServiceItem

Responses:

Status	Description & Headers	Response Schema
200	Success Header **ETag** (Opaque revision of the entire editable service aggregate (including fields/workflow). Store and send on next write.)	Ref: ServiceItem
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
412	Stale version: changed by another user, no write occurred	Ref: Problem
422	Validation error	Ref: Problem
428	Missing required If-Match precondition	Ref: Problem

[GET] /api/v1/services/{serviceItemUuid}/available-transitions

Operation ID: listServiceTransitions | **Tag:** Workflow | **Summary:** List currently authorized manual workflow transitions

Return currently authorized manual transitions from selected workflow instance. Do not hardcode legacy numeric statuses.

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	string (uuid)	-

Responses:

Status	Description & Headers	Response Schema
200	Success	array
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	Ref: Problem

[POST] /api/v1/services/{serviceItemUuid}/transitions

Operation ID: advanceServiceWorkflow | **Tag:** Workflow | **Summary:** Execute an atomic workflow state transition with If-Match guard

Requires If-Match; when destination is on_hold, mandatory holdType + holdReasonCode; commit service status, hold details and history in one DB transaction. Temporary lookup reason choices are for authorized test/demo environments only.

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	string (uuid)	-
`If-Match`	header	**Yes**	string	ETag of loaded service aggregate. Required on mutations. 412 if stale, 428 if absent.

Request Body: (Required: Yes) Schema: Ref: TransitionRequest

Responses:

Status	Description & Headers	Response Schema
200	Success Header **ETag** (New aggregate version after atomic transition)	Ref: TransitionResult
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
412	Stale version: changed by another user, no write occurred	Ref: Problem
422	Validation error	Ref: Problem
428	Missing required If-Match precondition	Ref: Problem

[GET] /api/v1/services/{serviceItemUuid}/history

Operation ID: getServiceHistory | **Tag:** Workflow | **Summary:** Get status and workflow transition history for a service instance

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	string (uuid)	-

Responses:

Status	Description & Headers	Response Schema
200	Success	array
400	Bad request / invalid field value	Ref: Problem
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	Ref: Problem

[GET] /api/v1/orders/{orderUuid}/eligible-billed-parties

Operation ID: listEligibleOrderItemBilledParties | **Tag:** Service Instances | **Summary:** List eligible service-level billed parties for one authorized order

Resolve order Doctor/default and Scanning Center within tenant and staff permission, return contact UUIDs for selection; do not return unrestricted contacts. Endpoint proposal until backend contract review.

Parameters:

Name	In	Required	Type	Description
`orderUuid`	path	**Yes**	string (uuid)	-

Responses:

Status	Description & Headers	Response Schema
200	Eligible billed parties	array
401	Not authenticated	-
403	Denied	-

Status	Description & Headers	Response Schema
404	Order not found or tenant-masked	-

[GET] /api/v1/catalog/hold-reasons

Operation ID: listEligibleHoldReasons | **Tag:** Catalog, Lookups | **Summary:** Return configured hold reasons; temporary synthetic choices are nonproduction only

Returns valid hold reasons filtered by holdType (customer or internal) for atomic on_hold workflow transitions (Q-B9).

Parameters:

Name	In	Required	Type	Description
`holdType`	query	**Yes**	Ref: HoldType	Required hold type filter (customer or internal)

Responses:

Status	Description & Headers	Response Schema
200	Configured hold reasons	array
401	Not authenticated	-
403	Denied	-

[GET] /api/v1/lookups/zero-charge-reasons

Operation ID: listZeroChargeReasons | **Tag:** Lookups | **Summary:** List active zero charge reasons

Returns active reasons required when an order item line total is $0.00 to satisfy DB constraint ck_order_items__zero_charge_has_reason (Q-B10).

Responses:

Status	Description & Headers	Response Schema
200	Success	array
401	Not authenticated	-
403	Denied	-

[GET] /api/v1/lookups/hold-reasons

Operation ID: listHoldReasons | **Tag:** Lookups | **Summary:** List active hold reasons (lookup alias for /api/v1/catalog/hold-reasons)

Returns valid reasons when placing an item on hold with customer or internal hold type; temporary synthetic choices are nonproduction only.

Parameters:

Name	In	Required	Type	Description
`holdType`	query	No	Ref: HoldType	Filter reasons by hold type (customer or internal)

Responses:

Status	Description & Headers	Response Schema
200	Success	array
401	Not authenticated	-
403	Denied	-

[GET] /api/v1/auth/me

Operation ID: getCurrentAuthContext | **Tag:** Auth | **Summary:** Get the current authenticated user and tenant authorization context

Responses:

Status	Description & Headers	Response Schema
200	Current authenticated user context	Ref: AuthMeResponse
401	Missing, invalid, expired, revoked, or idle-timed-out application session	Ref: AuthErrorResponse
403	Session tenant authorization was lost	Ref: AuthErrorResponse

[POST] /api/v1/auth/logout

Operation ID: logoutApplicationSession | **Tag:** Auth | **Summary:** Revoke the current application session

Requires a valid authenticated session. The handler safely resolves the server-owned session tenant before revoking the session and always returns 204 with no response body. Unknown or already-revoked tokens are safe no-ops.

Responses:

Status	Description & Headers	Response Schema
204	Session revoked; response has no body	-
401	Missing, invalid, expired, revoked, or idle-timed-out application session	Ref: AuthErrorResponse
403	Session tenant authorization was lost	Ref: AuthErrorResponse

[GET] /api/v1/iam/users

Operation ID: listIamUsers | **Tag:** IAM/Admin | **Summary:** List tenant-scoped employee identities for IAM administration

Responses:

Status	Description & Headers	Response Schema
200	Tenant-scoped employee identities and assigned roles	array
401	Authentication required	-
403	Missing iam.users.read capability or tenant authorization	-

[GET] /api/v1/iam/roles

Operation ID: listIamRoles | **Tag:** IAM/Admin | **Summary:** List tenant-scoped IAM roles and capabilities

Responses:

Status	Description & Headers	Response Schema
200	Tenant-scoped IAM roles	array
401	Authentication required	-
403	Missing iam.roles.read capability or tenant authorization	-

[POST] /api/v1/iam/users/{userId}/roles

Operation ID: assignIamUserRole | **Tag:** IAM/Admin | **Summary:** Assign a tenant role to an employee

Requires iam.assign_role. The role is resolved inside the authenticated session tenant. ADMIN-to-ADMIN delegation is explicitly not allowed by the current approved model.

Parameters:

Name	In	Required	Type	Description
`userId`	path	**Yes**	integer (int64)	-

Request Body: (Required: Yes) Schema: Ref: IamRoleAssignmentRequest

Responses:

Status	Description & Headers	Response Schema
200	Existing or newly created idempotent role assignment	Ref: IamRoleAssignmentResponse
400	Invalid userId or roleId	-
401	Authentication required	-
403	Missing iam.assign_role, invalid target/role tenant, or ADMIN delegation not allowed	-

Component Schemas

Schema: AuthLoginRequest (object)

Property	Type & Constraints	Req	Description
idToken	string	Yes	Google ID token used for authentication only; never treated as an authorization or tenant claim.

Schema: AuthorizedLoginResponse (object)

Property	Type & Constraints	Req	Description
accessStatus	string [enum: AUTHORIZED]	Yes	-
token	string	Yes	Opaque server-issued application session token.
expiresAt	string (date-time)	Yes	-

Schema: PendingAccessLoginResponse (object)

Property	Type & Constraints	Req	Description
accessStatus	string [enum: PENDING_ACCESS]	Yes	Authenticated identity has no valid tenant role assignment; no application session or token is issued.

Schema: AuthMeResponse (object)

Property	Type & Constraints	Req	Description
email	string (nullable)	Yes	-
userUuid	string (uuid) (nullable)	Yes	-
tenantSlug	string (nullable)	Yes	-
roleCodes	array	Yes	-
capabilities	array	Yes	-
isPendingAccess	boolean	Yes	Current capability-resolution Pending Access indicator; tenant authorization is separately revalidated by protected middleware.

Schema: AuthErrorResponse (object)

Property	Type & Constraints	Req	Description
error	Ref: Problem	Yes	-

Schema: IamRoleAssignmentRequest (object)

Property	Type & Constraints	Req	Description
roleId	integer (int64)	Yes	-

Schema: IamRoleAssignmentResponse (object)

Property	Type & Constraints	Req	Description
userId	integer (int64)	Yes	-
roleId	integer (int64)	Yes	-
roleCode	string	Yes	-
tenantId	integer (int64)	Yes	-
grantedAt	string (date-time)	Yes	-

Schema: IamUserSummary (object)

Property	Type & Constraints	Req	Description
userId	integer (int64)	Yes	-
email	string	Yes	-
state	string	Yes	-
hasEmployeeProfile	boolean	Yes	-
roles	array	Yes	-

Schema: IamAssignedRole (object)

Property	Type & Constraints	Req	Description
roleId	integer (int64)	Yes	-
roleCode	string	Yes	-

Schema: IamRoleSummary (object)

Property	Type & Constraints	Req	Description
roleId	integer (int64)	Yes	-
code	string	Yes	-
name	string	Yes	-
isSystem	boolean	Yes	-
capabilities	array	Yes	-

Schema: Problem (object)

Property	Type & Constraints	Req	Description
code	string	Yes	-
message	string	Yes	-
requestId	string	No	-
details	object	No	For ZERO_CHARGE_REASON_REQUIRED (422), details.field must equal zeroChargeReasonCode; other errors may include structured code-specific detail.

Schema: FormOption (object)

Property	Type & Constraints	Req	Description
code	string	**Yes**	-
label	string	**Yes**	-
active	boolean	No	-

Schema: CatalogService (object)

Property	Type & Constraints	Req	Description
uuid	string (uuid)	**Yes**	-
code	string	**Yes**	-
name	string	**Yes**	-
itemKind	string [enum: service]	**Yes**	-
isDigital	boolean	No	-
workflowCode	string (nullable)	No	-

Schema: CanonicalFieldType (string)

Data types allowed by V3 catalog.field_types.data_type (Q-B11), verify actual seeded code values; UI widgets such as radio vs select are presentation-only.

Allowed Values: text, number, boolean, date, single_select, multi_select, file, tooth_chart

Schema: FormValidation (object)

Known V3 validation keys (Q-B12); backend must review actual existing JSON configurations before freezing full key vocabulary.

Property	Type & Constraints	Req	Description
min	number	No	-
max	number	No	-
max_length	integer	No	-
pattern	string	No	-

Schema: FieldCondition (object)

Variants (oneOf): Ref: FieldConditionLeaf | Ref: FieldConditionAll | Ref: FieldConditionAny | Ref: FieldConditionNot

Schema: FieldConditionLeaf (object)

Exactly one of field/attribute; field codes resolved using tenant catalog; value vs values depends on op; enforce with application validator.

Property	Type & Constraints	Req	Description
`field`	string	No	-
`attribute`	string	No	-
`op`	string [enum: eq, ne, in, not_in, is_set,...]	**Yes**	-
`value`	any	No	-
`values`	array	No	-

Schema: FieldConditionAll (object)

Property	Type & Constraints	Req	Description
`all`	array	**Yes**	-

Schema: FieldConditionAny (object)

Property	Type & Constraints	Req	Description
`any`	array	**Yes**	-

Schema: FieldConditionNot (object)

Property	Type & Constraints	Req	Description
`not`	Ref: FieldCondition	**Yes**	-

Schema: ConditionalFieldRule (object)

Rule kind, ordered options precedence and conditional expression V3-derived (Q-B12); server evaluator authoritative.

Property	Type & Constraints	Req	Description
`ruleKind`	string [enum: visible, required, options]	**Yes**	-
`condition`	Ref: FieldCondition	**Yes**	-
`optionCodes`	array	No	-
`sortOrder`	integer	No	-
`active`	boolean	No	-

Schema: FormField (object)

Property	Type & Constraints	Req	Description
uuid	string (uuid)	**Yes**	-
code	string	**Yes**	-
label	string	**Yes**	-
type	Ref: CanonicalFieldType	**Yes**	-
required	boolean	**Yes**	-
defaultValue	string (nullable)	No	-
validation	Ref: FormValidation	No	-
conditionalRules	array	No	-
options	array	**Yes**	-

Schema: FormDefinition (object)

Property	Type & Constraints	Req	Description
service	Ref: CatalogService	**Yes**	-
fields	array	**Yes**	-
evidence	string	No	Indicates whether tenant runtime mapping is verified

Schema: FormAnswer (object)

Form answer model supporting unified optionCodes array for choice-based fields and natural polymorphic value for free-form inputs. Validation is performed from tenant catalog + source-evidenced rules, including conditional requiredness.

Property	Type & Constraints	Req	Description
fieldCode	string	**Yes**	Stable machine code of the form question (from catalog.fields.code)
value	oneOf: string | number | boolean	No	Natural polymorphic answer value for free-form inputs (text, number, boolean, date): string (text or ISO date YYYY-MM-DD), number, or boolean. Backend validates and maps to appropriate SQL column.
optionCodes	array	No	Unified option codes array for choice-based fields: single-select fields supply 0 or 1 element e.g. ["FORMLABS_3B"], multi-select fields supply 0..n elements e.g. ["UPPER", "LOWER"], and unselected/cleared fields supply [].

Schema: CreateServiceItem (object)

Service-level payer is backed by F07 V204 orders.order_items.billed_to_contact_id. Application resolves billedToContactUuid to crm.contacts.id under tenant-scoped transaction. Omit to select order default once; do not allow client to set provenance. Financial writes require runtime F07 gates. Every service instance independently retains billability and intended payer.

Property	Type & Constraints	Req	Description
`sellableItemUuid`	string (uuid)	**Yes**	-
`answers`	array	**Yes**	Submitted form answers. One entry per answered field; see FormAnswer.
`isRedo`	boolean [default: False]	No	-
`redoOfItemUuid`	string (uuid)	No	-
`serviceNotes`	string	No	-
`billedToContactUuid`	string (uuid)	No	Optional selected payer for THIS item; when omitted creation copies order default ONCE. Must be eligible for the specific order and tenant.
`isBillable`	boolean [default: True]	No	Independent service-instance billability (Q-B14); true = chargeable/billable, false = never invoiced per V3 column definition. Not the Doctor/Scanning Center payer selection; never map legacy toBeBilled to this flag. Changes require clerk permission and are financially locked once charged/invoiced.
`unitPrice`	string	No	Authorized CP-adjustable per-unit monetary price, decimal string (Q-B10). Backend supplies contract-based initial value, then accepts permitted clerk override without silently re-pricing it to contract amount. Enforce currency scale, audit and financial locks; do not trust unprivileged writers.
`quantity`	number [default: 1]	No	Selected demo workflows default to 1 (Q-B13). Do NOT interpret as universal integer-only or minimum 1: V3 supports >0 and up to 3 decimals (numeric(10,3)); item-specific policy pending.
`zeroChargeReasonCode`	string	No	Lookup code resolved server-side (Q-B10); mandatory for new lines whose final line_total equals zero, including discount/voucher, to satisfy DB constraint ck_order_items__zero_charge_has_reason.
`zeroChargeNote`	string	No	Free-text explanation when line_total is zero (max 255 chars).

Schema: PatchServiceItem (object)

Do not accept caller-supplied billedPartyOrigin. Reject per-item payer changes once a charge, invoice association or invoice line exists. Other monetary changes obey their own permissions and V3 financial locks (e.g. invoiced lines reject unitPrice, quantity, currency, and payer changes). Never expose arbitrary status or hold changes here instead of configured workflow actions (POST /transitions). Every service instance independently retains billability and intended payer.

Property	Type & Constraints	Req	Description
`answers`	array	No	Complete replacement list of form answers. One entry per answered field; see FormAnswer.
`isRedo`	boolean	No	-
`redoOfItemUuid`	string (uuid)	No	-
`serviceNotes`	string	No	-
`pendingReason`	string	No	-
`billedToContactUuid`	string (uuid)	No	Change intended payer for this single service instance only before any financial event; server writes billed_party_origin=user_selected and audits the change.
`isBillable`	boolean	No	Independent service-instance billability (Q-B14); true = chargeable/billable, false = never invoiced per V3 column definition. Not the Doctor/Scanning Center payer selection; never map legacy toBeBilled to this flag. Changes require clerk permission and are financially locked once charged/invoiced. This property changes billability only, not billedToContactUuid.
`unitPrice`	string	No	Authorized CP-adjustable per-unit monetary price, decimal string (Q-B10). Backend supplies contract-based initial value, then accepts permitted clerk override without silently re-pricing it to contract amount. Enforce currency scale, audit and financial locks; do not trust unprivileged writers.
`quantity`	number	No	Demo currently defaults to 1 (Q-B13); per-service validation pending; V3 permits positive decimals up to 3 places.
`zeroChargeReasonCode`	string (nullable)	No	Lookup code resolved server-side (Q-B10); mandatory when final line_total equals zero, including discount/voucher. Can be omitted or nullified if final total is nonzero.
`zeroChargeNote`	string	No	-

Schema: ServiceItem (object)

Each service instance has independent item-level billed party, billability, pricing breakdown, form answers, status and history.

Property	Type & Constraints	Req	Description
uuid	string (uuid)	**Yes**	-
orderUuid	string (uuid)	**Yes**	-
sellableItemUuid	string (uuid)	**Yes**	-
lineNumber	integer	**Yes**	-
status	string	**Yes**	-
isRedo	boolean	**Yes**	-
redoOfItemUuid	string (uuid) (nullable)	No	UUID of parent item this line remakes
isInvoiced	boolean	**Yes**	-
currencyCode	string	**Yes**	-
unitPrice	string	**Yes**	Decimal string, not floating point
fieldsAmount	string	**Yes**	Sum of answer surcharges recalculated by trigger trg_order_item_field_values__refresh_item
discountAmount	string	**Yes**	Applied discount amount
voucherAmount	string	**Yes**	Applied voucher amount
lineTotal	string	**Yes**	Final line amount (unitPrice * quantity + fieldsAmount - discountAmount - voucherAmount)
canEditPrice	boolean	**Yes**	Backend-derived permission hint for authorized CP clerk (Q-B10); API/DB enforce.
answers	array	**Yes**	Form answers for this service instance. One entry per answered field; see FormAnswer.
serviceNotes	string	No	-
billedToContactUuid	string (uuid)	**Yes**	Authoritative item-level intended payer (crm.contacts.uuid), not orders.orders.contact_id or charges.contact_id.
billedPartyOrigin	Ref: BilledPartyOrigin	**Yes**	-
canEditBilledParty	boolean	**Yes**	Server-calculated permission and financial-lock hint; backend always enforces mutation rules.
isBillable	boolean [default: True]	**Yes**	Independent service-instance billability (Q-B14); true = chargeable/billable, false = never invoiced per V3 column definition. Read persisted orders.order_items.is_billable, independent from billedToContactUuid.
canEditBillability	boolean	**Yes**	Server-derived hint for authorized internal clerk (Q-B14); actual backend validation and financial lock remain mandatory.
quantity	number [default: 1]	**Yes**	Stored V3 quantity (numeric(10,3)); demo default 1 is not a general integer restriction (Q-B13).
pendingReason	string (nullable)	No	Reason the item is pending information
holdType	Ref: HoldType	No	Active hold type when status is on_hold
holdReasonCode	string (nullable)	No	Active hold reason code when status is on_hold
zeroChargeReasonCode	string (nullable)	No	Code explaining why line_total is zero
zeroChargeNote	string (nullable)	No	-
etag	string	**Yes**	Optional body convenience only (Q-B8); authoritative version is ETag response header. Backend must version entire editable aggregate.
createdAt	string (date-time)	**Yes**	-
updatedAt	string (date-time)	**Yes**	-

Schema: AvailableTransition (object)

Property	Type & Constraints	Req	Description
transitionId	integer (int64)	**Yes**	-
fromCode	string	**Yes**	-
toCode	string	**Yes**	-
label	string	No	-
requiresCapability	string	No	-
guard	object	No	-

Schema: TransitionRequest (object)

One atomic transition (Q-B9): when destination is on_hold, holdType and holdReasonCode are required; set status, hold type/reason, and workflow history together. Leaving hold clears active details while retaining history. Server validates against actual transition target.

Property	Type & Constraints	Req	Description
transitionId	integer (int64)	**Yes**	-
comment	string	No	-
holdType	Ref: HoldType	No	-
holdReasonCode	string	No	Required alongside holdType for EVERY new transition into on_hold (Q-B9), resolved to orders.order_items.hold_reason_id from eligible lookup. Must be absent/handled per workflow policy on transitions outside on_hold.

Schema: TransitionResult (object)

Property	Type & Constraints	Req	Description
serviceItemUuid	string (uuid)	**Yes**	-
fromCode	string	**Yes**	-
toCode	string	**Yes**	-
changedAt	string (date-time)	**Yes**	-
actorUserUuid	string (uuid)	No	-
holdType	Ref: HoldType	No	-
holdReasonCode	string	No	-

Schema: HistoryEvent (object)

Property	Type & Constraints	Req	Description
fromCode	string	No	-
toCode	string	No	-
actorUserUuid	string (uuid)	No	-
occurredAt	string (date-time)	No	-
comment	string	No	-

Schema: BilledPartyChoice (object)

Eligible contacts derived from selected order doctor/default contact and scanning center. Not an unrestricted tenant contact list.

Property	Type & Constraints	Req	Description
`contactUuid`	string (uuid)	**Yes**	Public identifier for crm.contacts; resolve bigint ID on backend only within trusted tenant scope
`role`	string [enum: doctor, scanning_center]	**Yes**	Choice role in this specific order, not global contact category
`label`	string	**Yes**	Approved display label; avoid PHI

Schema: BilledPartyOrigin (string)

Read-only migration/user-action provenance. The unresolved_legacy state is prohibited in authoritative completed ETL.

Allowed Values: legacy_explicit, order_default, manual_resolution, user_selected, unresolved_legacy

Schema: HoldType (string)

Hold category when placing an order item on_hold (customer vs internal).

Allowed Values: customer, internal

Schema: HoldReasonOption (object)

Approved catalog rows once available; current demo temporary synthetic nonproduction choices must be explicitly labelled (Q-B9). Never invent actual historical reason values.

Property	Type & Constraints	Req	Description
`code`	string	**Yes**	Unique code in core.lookup_values (or SYNTHETIC_* in non-prod demo)
`label`	string	**Yes**	Human-readable label
`isTemporary`	boolean	**Yes**	True when option is a temporary synthetic demo choice (never production/historical authority)
`holdType`	Ref: HoldType	No	-
`active`	boolean	No	-

Schema: ZeroChargeReason (object)

Property	Type & Constraints	Req	Description
`code`	string	**Yes**	Unique code in core.lookup_values
`label`	string	**Yes**	Human-readable label (e.g., Redo / Remake, Promotional Courtesy, Voucher Full Coverage)
`description`	string (nullable)	No	-
`active`	boolean	**Yes**	-